

FailSafe AI

Student Success Intelligence Platform

Submitted for:

Bharat Academix CodeQuest – Round 1

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The Problem

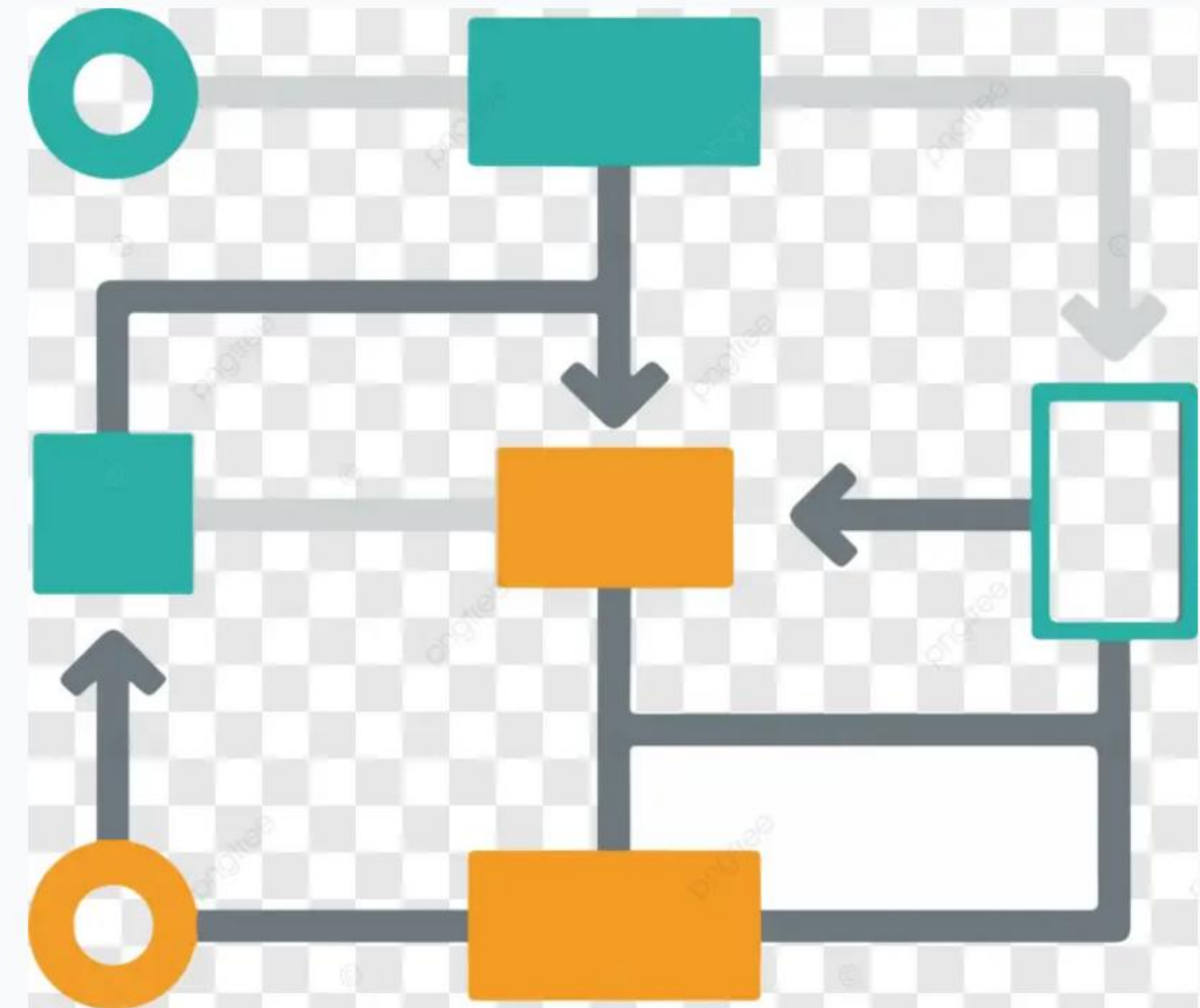
- ⚠ **Data without Guidance:** Students receive raw marks and attendance percentages but lack personalized, actionable guidance on how to improve.
- ⚠ **Late Career Preparation:** Resume building, skill acquisition, and career preparation often begin too late in the academic lifecycle, missing critical early opportunities.
- ⚠ **Unclear Learning Paths:** Many students struggle to identify which specific skills to learn next to bridge the gap between academia and industry requirements.
- ⚠ **Reactive Support:** Faculty currently have limited tools for proactive academic support, usually intervening only after a student has already underperformed.

Proposed Solution

FailSafe AI is an AI-powered decision-support platform designed to transform raw academic indicators into a structured roadmap for success.

- Analyzes core academic indicators (attendance, marks, assignments).
- Generates a comprehensive Academic Health Score instead of just GPA.
- Explains performance drops and identifies improvement opportunities.
- Recommends semester-specific, personalized learning plans.
- Provides actionable steps for internships and placement readiness.

$$\text{Academic Health Score} = \sum_{i=1}^n w_i \cdot \text{metric}_i$$



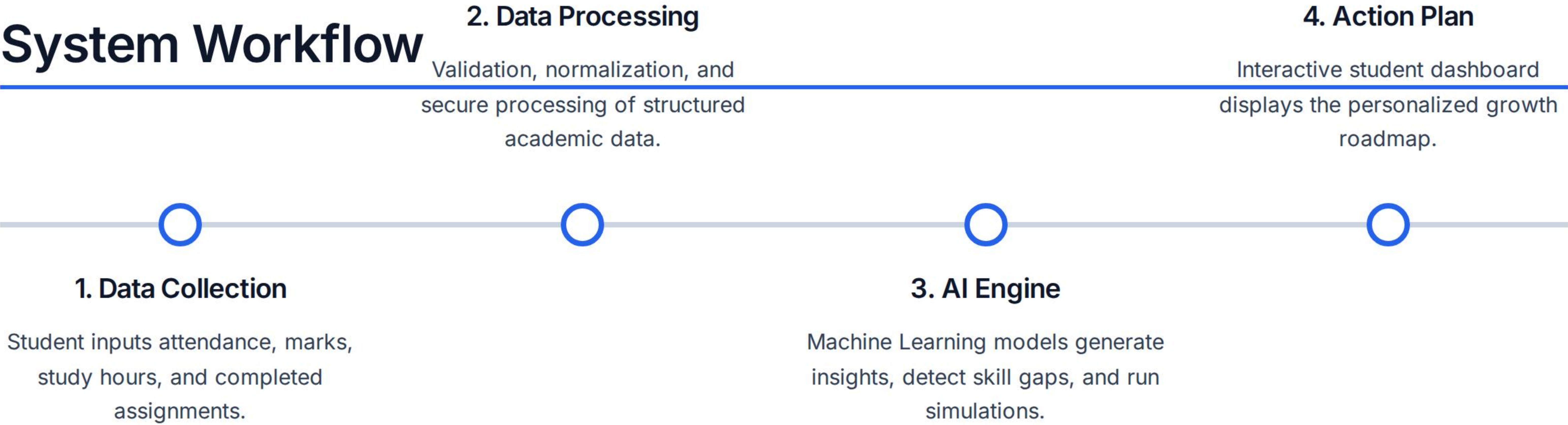
Key Features

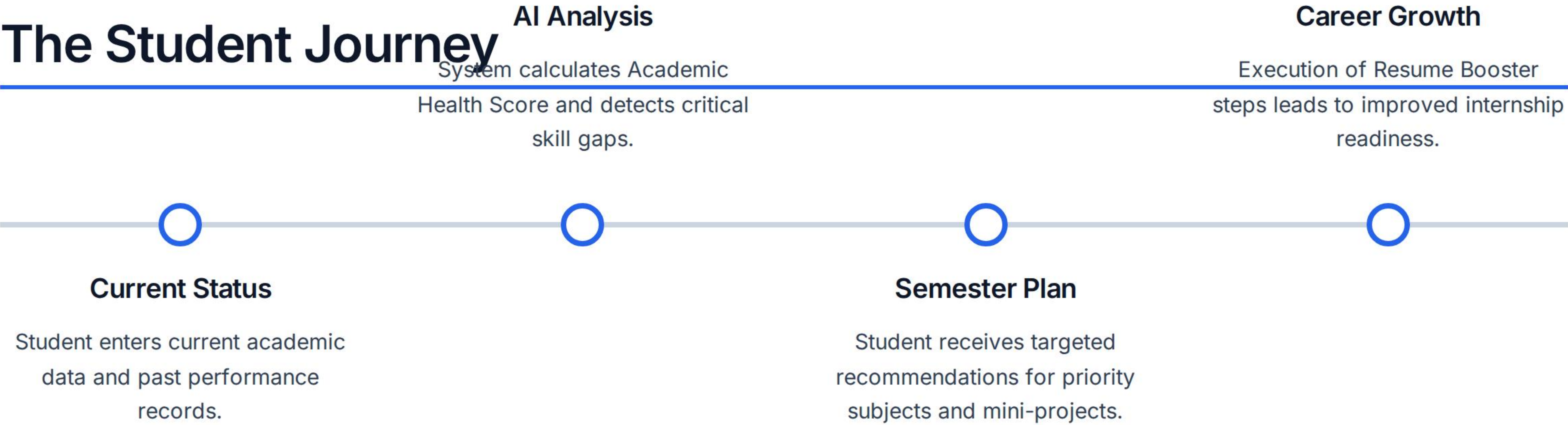
-  **Academic Health Score:** Holistic metric of performance.
-  **Recovery Planner:** Steps to bounce back from low grades.
-  **Academic Digital Twin:** Virtual profile of student capabilities.
-  **Skill Gap Analyzer:** Map current skills against job needs.
-  **Career Readiness Dashboard:** Track internship preparedness.
-  **Explainable Insights:** Understand exactly "why" a score changed.
-  **What-If Simulator:** Predict impact of future study habits.
-  **Semester Intelligence Engine:** Targeted focus areas per term.
-  **Resume Booster:** Concrete steps to enhance employability.
-  **Faculty Analytics Dashboard:** Class-wide trends and early warnings.

Why FailSafe AI Stands Out

Feature / Aspect	Existing Student Systems	FailSafe AI
Data Interpretation	Show raw attendance and marks only	Provides an Academic Health Score with visual analytics
Support Approach	Reactive support after problems occur	Proactive, personalized improvement recommendations
Academic Planning	General syllabus mapping	Semester Intelligence Engine with targeted suggestions
Career Integration	No integrated career guidance	Identifies skill gaps and offers Resume Booster guidance
Faculty Tools	Basic reporting spreadsheets	Interactive class-level analytics and early warnings

💡 "From data collection to data-driven decisions. FailSafe AI complements existing academic systems by transforming raw data into structured growth plans."





Stakeholder Impact



Students

Gain personalized guidance, structured academic planning, focused skill development, and robust preparation for internships and placements.



Faculty

Access clear performance insights and class-wide analytics, enabling early intervention and data-driven mentoring for at-risk students.



Institution

Benefit from overall improved academic outcomes, better placement records, and a modernized, supportive educational ecosystem.

Semester Intelligence Engine

▼ Input Selection

The student initiates the engine by selecting their current academic context:

- **Department:** e.g., AI & Data Science
- **Year of Study:** e.g., 2nd Year
- **Current Semester:** e.g., Semester 4
- **Career Goal:** e.g., Data Engineer

💡 AI Recommendations

The engine generates a structured, branching roadmap:

- **Priority Subjects:** Identify high-weightage courses.
- **Skills to Learn:** SQL, Python Data Structures.
- **Certifications:** AWS Cloud Practitioner.
- **Mini-Projects:** Real-time API data extraction.
- **Internship Prep:** Resume updates & interview prep.

System Architecture



2. Python Core

Data validation, cleaning, and preprocessing using Pandas and NumPy.

4. Dashboard UI

Streamlit web application renders interactive Plotly charts and insights.



1. Data Inputs

Structured academic CSV/JSON files, user metrics, and manual entry.



3. ML Engine

Scikit-learn models run predictive analysis and generate recommendations.



Technology Stack

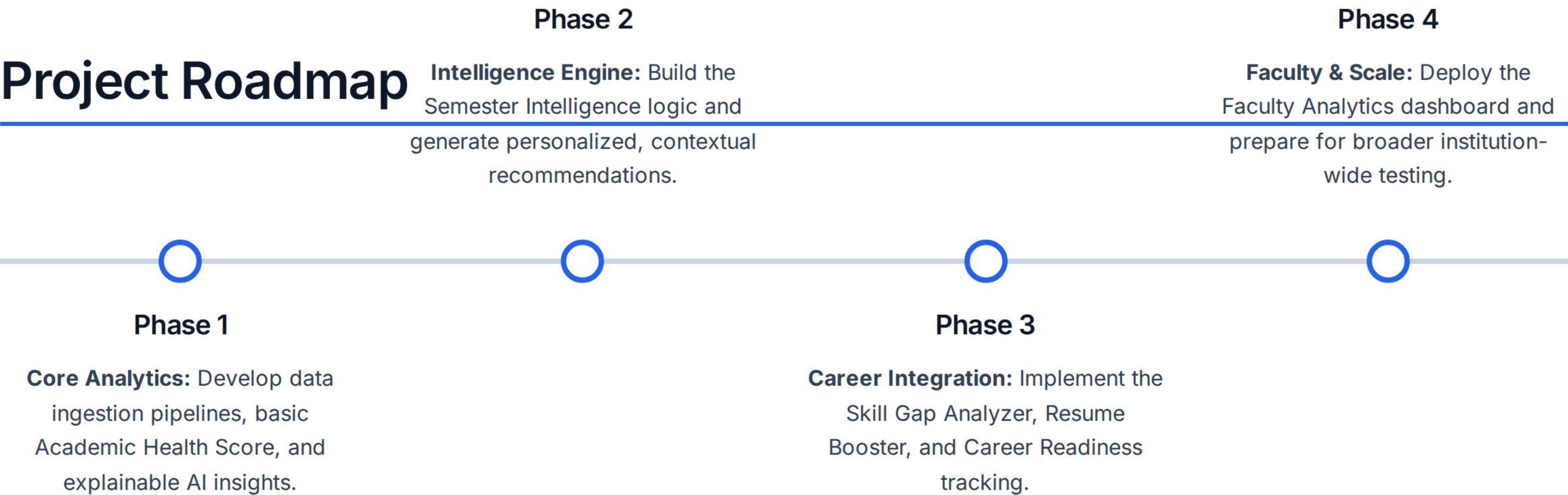
Application & Data Layer

- **Python:** Core programming language for logic and data handling.
- **Streamlit:** Rapid frontend framework for creating the interactive dashboard UI.
- **Pandas:** High-performance data manipulation and analysis.
- **NumPy:** Fundamental package for scientific computing and matrix operations.

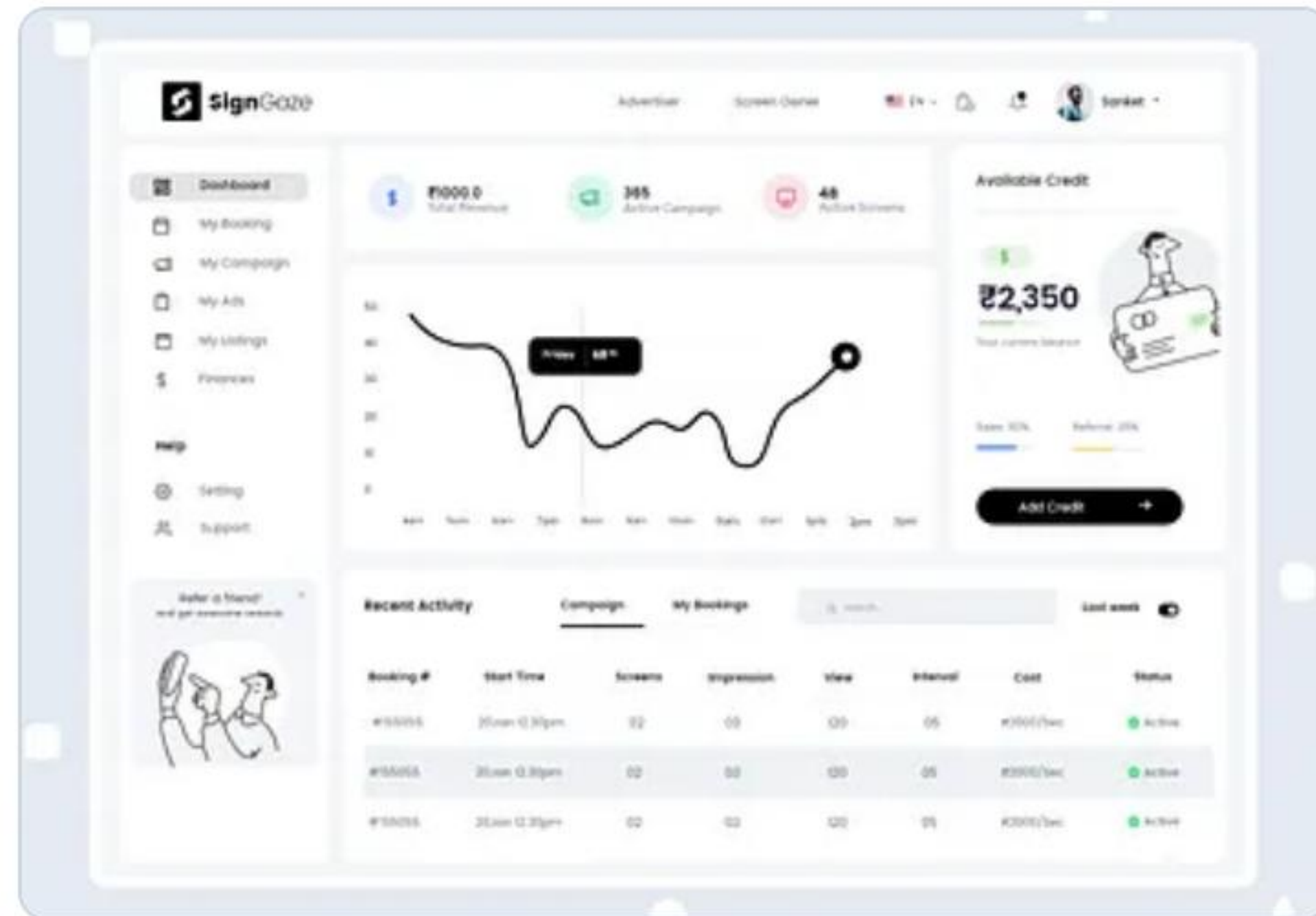
Analytics & Machine Learning

- **Scikit-learn:** Library for building predictive models (classification, regression).
- **Plotly:** Interactive, publication-quality graphing library for data visualization.
- **Pickle:** Serialization library for efficient model storage and loading.
- **Recommendation Engine:** Custom algorithmic logic for mapping skills to career goals.

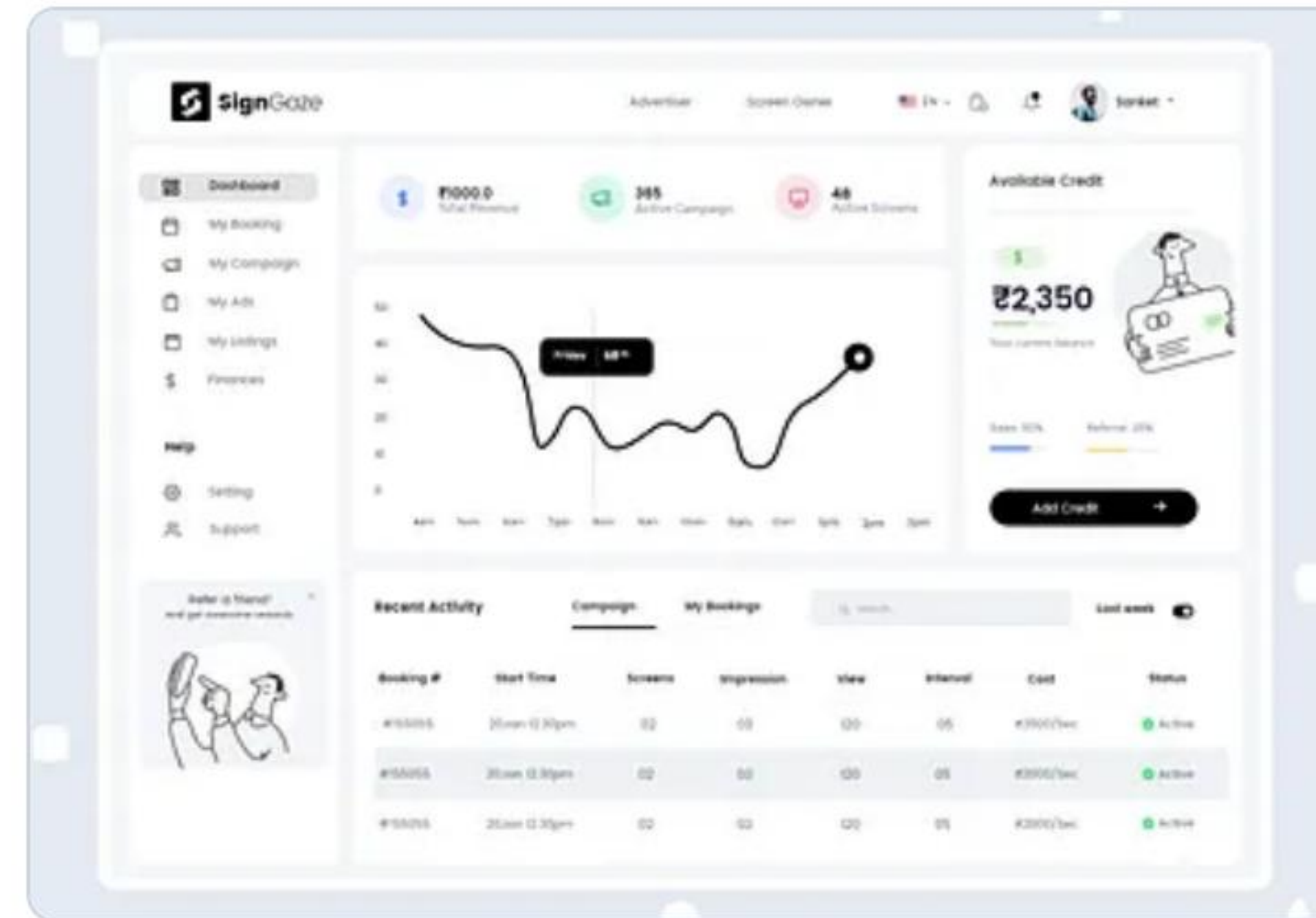
Project Roadmap



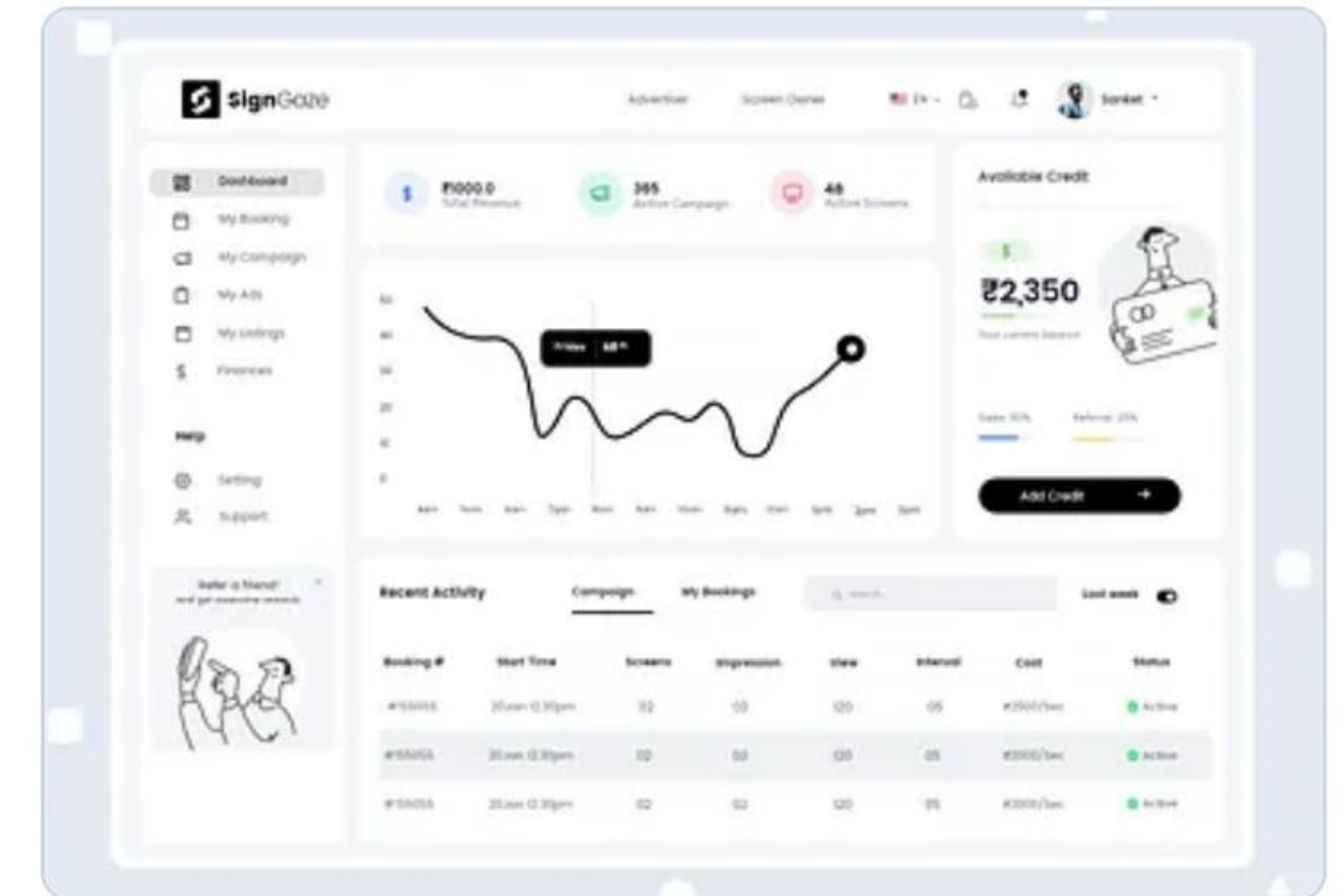
Concept Dashboards



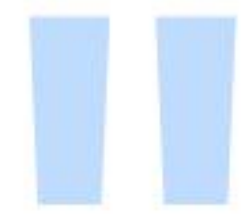
Student Dashboard
Health Score & Overview



Semester Intelligence
Subject & Skill Recommendations



Career Readiness
Resume Booster & Skill Gaps



**FailSafe AI empowers students with
personalized academic intelligence,
helping them make smarter decisions
and build successful careers.**



From Data to Direction. From Potential to Success.

Image Sources



Thumbnail
for
pngtree.com

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